

3 (a) (i) resultant force is zero	B1	
weight of plank + weight of man = $F_A + F_B$ or $200 \text{ (N)} + 880 \text{ (N)}$ or $1080 = F_A + F_B$	B1	[2]
(ii) principle of moments used	C1	
(anticlockwise moments) $F_B \times 5.0$	C1	
(clockwise moments) $880 \times 0.5 + 200 \times 2.5$	C1	
$F_B = (440 + 500)/5.0 = 188 \text{ N}$	A1	[4]
(b) straight line with positive gradient (allow freehand)	M1	
start point (0, 100)	A1	
finish point (5, 980)	A1	[3]

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